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THE LEGACY OF ALAN TURING

By Peter Love

Alan Turing: Life and Legacy of a Great Thinker, Christof Teuscher (ed.), Springer Verlag, 2004, US\$69.95.

THE ENGINEERING PROBLEM OF PRODUCING VARIOUS MACHINES FOR VARIOUS JOBS IS REPLACED BY THE OFFICE WORK OF ‘PROGRAMMING’ THE UNIVERSAL MACHINE TO DO THESE JOBS.” — ALAN TURING, 1948

It is difficult to overstate the impact of Alan Turing’s statement (above) on modern life. His 1937 work, “On Computable Numbers with an Application to the Entscheidungsproblem,” in which he placed the concept of computation on sound footing by defining Turing machines and distinguishing between hardware and software through the concept of a universal machine, stands as one of the 20th century’s intellectual triumphs.

Alan Turing: Life and Legacy of a Great Thinker, a collection of articles from speakers at Turing Day 2002 in Lausanne (commemorating Turing’s 90th birthday), together with other contributions, represents a solid, if occasionally eclectic, overview of Turing’s life, his work, the questions that motivated his research, and those that arose from it. Editor Christof Teuscher divides the volume into five sections: biography, mathematical logic, artificial intelligence, the Enigma machine, and a final chapter on Turing’s less well-known ideas regarding phyllotaxis and connectionism.

Turing’s Thesis

Turing’s thesis, which states that a Turing machine can capture any algo-

rithmic action, constitutes our deepest understanding of what the phrase “to compute” actually means. In addition, Turing’s universality thesis, which states that Turing machines can simulate any class of effective devices for computing, forms the basis of the division of the modern computing industry into hardware and software. Were we ignorant of the ideas of universal computation, it is entirely possible that we would have a glorified electronic typewriter sitting on our desks next to a special-purpose machine for sending text and documents and in front of a specific machine for simulating fluid mechanics.

However, alarm bells should be ringing for the more rigorously minded reader by now. What, exactly, is a “thesis”? Is it a theorem, a conjecture, or something else? This small chink in the armor of computing theory remains open, not because the Turing machine concept is unclear, but because our idea of an “algorithmic action” or “class of effective devices for computing” resists precise definition. A recurring idea in *Alan Turing* is that Turing used precise abstract concepts, such as the Turing machine and the Turing test, to attack

questions that otherwise evade definition. The enduring ability of Turing’s work to stimulate debate is well reflected in this volume.

Mathematical Logic

In the mathematical logic section, Michael Beeson of San Jose State University takes us headfirst into Turing’s motivation for his 1937 work in a tough but comprehensive overview of his negative solution of the Entscheidungsproblem and subsequent efforts to mechanize parts of mathematics in spite of the impossibility of mechanizing it all. In his description of the difficulties of mechanizing even some subset of mathematical activity, Beeson provides a great exposition of the power of Turing’s result and the difficulty of sidestepping it even slightly.

In spite of these difficulties, there remain efforts to define computational theories that go beyond the Turing model’s capabilities. This activity is dubbed *hypercomputation*, and “Hypercomputational Models” by Mike Stannett of University of Sheffield, and “Turing’s Ideas and Models of Computation” by Eugene Eberbach, Dina Goldin, and Peter Wegner, expound and characterize the various efforts to date. Such efforts rely on infinite resources, precision, and time. Of the three, infinite precision seems to be the most contentious. Turing was well aware of the problems associated with allowing oneself infinitely precise specification of a computing machine. When choosing a finite alphabet for his ma-

chine, Turing cited the justification that infinite alphabets would have characters resembling each other arbitrarily closely, which would lead to confusion.

Martin Davis of New York University gives a rebuttal of hypercomputation, well summarized by his conclusion that hypercomputational systems give noncomputable output only when noncomputable input is allowed. This section's for-and-against nature is entertaining as well as informative, leaving the reader with a better understanding of the limitations of algorithmic actions than would a simple reiteration of Turing's proof.

Both Stannett and Eberbach raise the possibility of using infinite time to go beyond the capabilities of Turing machines. This relies on the existence of Malament-Hogarth spacetimes, in which there is a path of infinite length starting at point p , at every point on which an observer can signal to an observer at point q , which can be reached from p in finite proper time. We can then launch a Turing machine down the infinite path with instructions to signal its results to q . We then travel to q and pick up our answer. The Turing machine has infinite time available to compute in, but we obtain our answer in finite time. This is a neat, if slightly impractical, evasion of the limitation of Turing machines' actions to finite time. Clearly one would like to find some good physical reason to exclude such spacetimes (just as spacetimes allowing time travel and warp drives are excluded). Unfortunately this remains open, although speculation exists that such spacetimes are "unstable" in some appropriate sense and so should not be observed in practice. It would indeed be remarkable if the consideration of mechanizing the process of

computing on squared paper could be intimately connected with deep properties of space and time.

Of course, many have speculated, most notably David Deutsch and Stephen Wolfram, that computability does indeed have something deep to teach us about physics. In Christopher Timpson's excellent article, he examines and rejects several of Deutsch's claims, in particular Deutsch's claim in his book *The Fabric of Reality* that mathematical truth is contingent on physical law, which receives a strongly critical treatment. Timpson clearly states the basic observation that a physical system can only be said to compute once a human gives the evolution of the system meaning. This remark is particularly apt in the context of quantum computing. If a nuclear magnetic resonance machine puts a system of nuclear spins through a certain coherent evolution, in what sense are the nuclear spins "computing"?

To quote Timpson, we must give "...a specification of what it is for a physical object to compute, to give a mathematical meaning to the possible evolutions of physical states." Put simply, every time we use the word "computation," we implicitly mean "computation by humans." Thanks to Turing, we can automate some of our computations by encoding them in a finitely specified physical process, but they remain our computations. It is the nature of humans' mental processes in this picture that remains open to debate.

Throughout the book's section on both mathematical logic and artificial intelligence (discussed later) runs the question of whether conscious human thought is algorithmic, whether algorithmic action can flawlessly imitate the result of conscious thought, and whether these two questions are distinct. I found that Timpson and Diane

Proudfoot's contributions to this volume shed new light on this debate by expounding the interplay of Turing's ideas with those of Ludwig Wittgenstein. Turing and Wittgenstein met frequently while peers at Cambridge, but Wittgenstein retained his view that asking whether machines can think is like asking if "the number three has a colour."

Proudfoot develops Wittgenstein's distinction between following a rule and acting in accordance with one. If this sounds like a distinction without a difference, Proudfoot gives the example of a native English speaker who follows the language's grammar rules, compared to a non-English speaker who has memorized a set of grammatically perfect sentences. Both speakers obey the rules of English grammar, but only the native speaker can be said to be following the rules.

Artificial Intelligence

Many of the contributions to *Alan Turing* skewer themselves on the horns of the dilemma most clearly posed by Roger Penrose. Either everything, including physical law, is the result of algorithmic action, in which case we are all in principle Turing machines, strong AI will ultimately succeed, and we have some odd delusions about the foundations of mathematical truth. Or, some aspect in the physical universe is not the result of algorithmic action; hence, even algorithmic actions can perform super-Turing feats by interacting with nonalgorithmic entities. However, in the second case, once we admit noncomputable action in nature, it requires no great leap to suppose that such action might be present in human thought, which will then elude action of any human-constructed computing device. While these issues will undoubtedly stimulate debate for decades

to come, this volume succeeds in doing more than simply plough a well-known furrow one more time.

Enigma Code Machine

No less remarkable than Turing's work were the major events of his life. If North American mathematicians and physicists were drawn to the Manhattan Project, their British counterparts went to Bletchley Park to work on German enigma code decryption. In September 1939, Turing went to Bletchley Park, where he became indispensable in decrypting German signals intelligence there. Winston Churchill considered the intelligence available from Bletchley vital; he referred to the codebreakers as "The geese who laid the golden eggs and never cackled." Andrew Hodges' excellent articles, "Alan Turing: An Introductory Biography" and "What Would Alan Turing Have Done After 1954?," provide a vignette of his definitive Turing biography and a speculative look on what Turing would have achieved had he lived past 1954.

The section on the Enigma machine fills in the gaps around the several-times-told story of Turing's contribution toward breaking the Enigma code. The story of the Polish decryption of the Enigma code prior to 1939 is told in "The Polish Brains Behind the Breaking of the Enigma Code Before and During the Second World War," by Elisabeth Rakus-Andersson of the Blekinge Institute of Technology, as well as a technically detailed account of Turing's work at Bletchley Park and a brief description of Turing's role as US-UK liaison on Enigma work.

Almost Forgotten Ideas

The final chapter on Turing's "almost forgotten ideas" gives some fascinating results of Turing's simulations of reac-

tion diffusion systems, including figures of what must be the first scientific visualizations ever produced. Teuscher, ever the modest editor, places his contribution at the end of the collection, in which he discusses the relationship of Turing's "unorganized machines" with modern artificial neural networks.

While I began by saying it is difficult to overstate the impact of Turing's work to date, it is easy to make wild extrapolations of future developments in computing. Ray Kurzweil's essay in this collection takes no prisoners in this regard, and is an enormously enjoyable romp through the implications of continued exponential growth of technological capacity. I find it difficult to take his vision of the future, in which we are all partially or completely assimilated by nanorobotic elements (rather like the Borg in *Star Trek*), entirely seriously. However, as Niels Bohr remarked, "Prediction is difficult, especially about the future," and perhaps I am merely exposing myself to future mockery by our (suitably enhanced) descendants.

Alan Turing: Life and Legacy of a Great Thinker is not an introductory text, although the articles are accessible to a general scientific audience. Its copious citations provide a good guide to the introductory works in the literature such as Martin Davis' *Computability and Unsolvability* and Andrew Hodges' biography of Turing, *Alan Turing: The Enigma*. Readers who wish to go a little beyond such introductory presentations will find this text particularly interesting.

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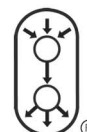
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